MIL-S-46376(MU)
10 September 1963

MILITARY SPECIFICATION

SIGHT ANTIOSCILLATION: XM55

1. SCOPE

1.1 This specification covers one type of Binocular viewing device known as Sight, Antioscillation: XM55.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-O-13820	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

IEL-8587675	List of Inspection Equipment Numbers for Sight Antioscillation: XM55
F8587675	Sight Antioscillation: XM55

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

PACKAGING DATA SHEET

MIL-P-14232/8587675 Packaging of Sight, Antioscillation XM55

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Construction.- The sight shall be manufactured in accordance with Drawing F8587675 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical, mechanical characteristics, performance and permanence to assure inherent quality and performance capabilities.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Cleanliness

3.3.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperature.- With the pip pin in position locking the gimbal and the inner frame to the outer frame, the sight shall be exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit. Upon return to standard ambient temperature (plus 60 to 90 degrees Fahrenheit) the sight shall show no evidence of physical failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

3.4.2 Operating temperature.- With the sight exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit the following requirements shall be met.

(a) The pip pin shall be removed and placed in its storage area.

(b) The gimbal assembly shall be movable about horizontal and vertical axis.

(c) The headrest shall be operable and shall lock in detent.

(d) The interpupillary adjustment shall be operable.

3.4.3 Post environmental.- Subsequent to meeting the requirements of 3.4.1 and 3.4.2 the sight shall meet the requirements of 3.5 to 3.12 inclusive.

3.5 Oscillation.- With the sight completely assembled and final adjustments of the counter weights and dashpots completed, the headrest in the position closest to the frame, the interpupillary distance recorded, the pip pin removed and placed in its storage area, the sight shall be oscillated. During oscillation the sight shall meet the requirements of Table I.

TABLE I

Drive to shake Table in R.P.M.	Frequency of Vibration in cps	Horizontal and Vertical Component - Maximum Transmissability (T) value
40		3.1
50		3.2
60	1.0	3.3
70		3.2
80		2.6
120	2.0	1.0
150		0.5
180	3.0	0.4
350	5.5	0.2
350 Through 700		0.2

(a) Transmissability of 1.0 is established with a rigid sight, (unit with pip pin in place).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

(b) At one cps (cycle per second) the sight shall oscillate at resonance.

(c) With the pip pin removed, $T = \frac{\text{Amplitude of Telescopes}}{\text{Amplitude of Shake Table}}$

(d) Value in Table I for horizontal and vertical components shall be an average value.

3.6 Collimation.- Subsequent to meeting the requirements of 3.5 the pip pin shall be placed in position to lock the gimbal. The interpupillary distance shall not have changed. The emergent lines of sight to a target shall be horizontal, and the images of an infinity target formed by the two telescopes shall be parallel to each other within one degree. Dipvergence shall not exceed 12 minutes of arc, total. Divergence shall not exceed 24 minutes of arc nor converge 12 minutes of arc all in the apparent field. These conditions shall be met at interpupillary settings of 58 and 72 millimeters.

3.7 Cleanliness.- The sight shall meet the cleanliness requirements of Specification MIL-O-13830 and there shall be no more than 3 particles of dirt visible on the surfaces of the optics of each telescope. Each particle of dirt shall not be in excess of the area equivalent to a size 5 dig. There shall be no other foreign matter, dirt or smears on the surfaces of the optics of each telescope which interferes with observation or adversely affects image quality.

3.8 Interpupillary adjustment.- Rotation of the interpupillary adjustment knob shall move the left and right telescope to position the centers of the exit pupils over a range of 58 to 72 millimeters.

3.9 Eyepiece focus.- When viewing a target at 2000 meters, the eyepiece focus of each telescope shall be fixed between the limits of minus 0.75 and minus 1.0 diopter.

3.10 Magnification.- The magnification of each telescope shall be six power plus or minus two percent. The magnification of the two telescopes of any one sight shall not differ from each other by more than two percent.

3.11 Resolution.- The resolving power of each telescope, on the optical axis, shall be five seconds of arc or less.

3.12 Image tilt.- The image of a plumb line target formed by the optical system of each telescope shall be vertical within one degree and the images of the plumb lines formed by the two telescopes shall be parallel to each other within thirty minutes of arc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Presentation of Lots and Drawing of Samples" specified in Standard MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table III-A with
Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None

MAJOR: AQL 1.0 Percent Defective		Requirement	Test Procedure
101.	Incorrect oscillation	3.5	4.5.3
102.	Pre-collimation error	3.6	4.5.2
103.	Excessive movement	3.6	4.5.3.1
104.	Improper cleanliness	3.7	4.5.3.7
105.	Magnification	3.10	4.5.3.6
106.	Insufficient eyepiece focus	3.9	4.5.3.2
107.	Insufficient resolution	3.11	4.5.3.4
108.	Insufficient image tilt	3.12	4.5.3.3
MINOR: AQL 4.0% Defective			
201.	Improper interpupillary Adjustment	3.8	4.5.3.5

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Disposition of Non-conforming Product" specified in Standard MIL-STD-105.

4.3 Control tests.- Three sights shall be selected at random from each 75 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Material	3.1	Par 3.6 of MIL-I-45208
Construction	3.2	Visual - drawing
General Specifications	3.3	Visual MIL-F-13926 MIL-O-13830
Environmental	3.4	4.5.1

4.3.1 Control test failure.- Should the test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL-I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the List of Inspection Equipment Numbers, IEL-8587675. Alternate design may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 Test methods and procedures.

4.5.1 Storage temperature.- This test shall be performed as outlined in Specification MIL-F-13926, however, the temperatures specified in 3.4.1 shall apply. During the test the pip pin shall be inserted in position as specified in 3.4.1. Upon thermal stabilization at room temperature (60 to 90 degrees Fahrenheit) the sight shall be examined for the requirement of 3.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

4.5.1.2 Operating temperature.- The equipment shall be placed in the test chamber and the temperatures reduced gradually to the specified condition outlined in 3.4.2 until thermal stabilization is reached. The sight shall then be tested for conformance to the requirements outlined in 3.4.2. The temperature shall be raised gradually to the specified condition outlined in 3.4.2 until thermal stabilization is reached. Then the sight shall again be tested for conformance to the requirement of 3.4.2.

4.5.2 Pre-collimation.- Pre-collimation shall be performed prior to the oscillation test using a device conforming to the accuracy specified on Drawing F8205654. Insert the sight into the holding medium and secure in position by the means provided. Set the testing medium and sight to an interpupillary setting of 58 MM. Sight through the viewing medium of the testing device and observe coincidence between the optical axis of the testing medium and the sight. Dipvergence, divergence and convergence shall be within the limits of 3.6 as measured by the viewing medium of the testing device. Record these three positions.

4.5.2.1 Repeat this procedure at the other interpupillary setting (72 MM). Dipvergence and divergence and convergence measured as specified in 4.5.2 shall be within the limits outlined in 3.6. Record these positions.

4.5.3 Oscillation.- After pre-collimating the sight, assemble the sight to the holding medium used to support the sight during oscillation test and proceed as outlined in the set up and operational instruction sheet provided with the oscillation fixture. After completion of this test remove the sight from the fixture.

4.5.3.1 Collimation.- With the sight again secured to the test medium repeat the procedure outlined in 4.5.2 and 4.5.2.1. The optical axis of the sight shall not move from the pre-collimated positions outlined in 4.5.2 and 4.5.2.1 by more than that amount specified in 3.6.

4.5.3.2 Eyepiece focus.- With the sight positioned in the testing device outlined in 4.5.2 each telescope assembly of each sight shall produce a clear sharp image of the projected target (reticle of the testing medium) within the limits of 3.9. Without distrubing the eyepiece setting of the viewing medium, focus either the eyepiece and reticle assembly or objective assembly until the image produced by the sight is in sharpest focus. The diopter setting as read on the viewing medium shall be within the limits of 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

4.5.3.3 Image tilt.- With the sight positioned in the test medium as specified in 4.5.2 the images of both vertical target lines (reticles of the testing mediums) shall be vertical with the corresponding centerline of the viewing mediums (reticle line) within the limits specified in 3.12. Also the angular displacement between the vertical target line (reticle line) projected through each telescope assembly of the sight, when observed by the viewing medium shall be within the parallel limits outlined in 3.12.

4.5.3.4 Resolution.- With the sight positioned as specified in 4.5.2 the image of the test bar patterns located about the optical axis of each reticle of the testing medium shall be positioned at or near the sight's optical axis by positioning the sights optical assembly within the frame of the sight after removing the pip pin. The optical assemblies will be retained in this position by means of blocks or shims placed against the inner frame. When this position has been obtained the test bar pattern shall be resolved when viewed through the viewing medium of the testing device. The test bar patterns are considered resolved when the observer can distinquish the direction in which the pattern lies in order to conform to the requirements of 3.11.

4.5.3.5 Interpupillary adjustment.- The amount of interpupillary movement can be measured by means of standard measurement equipment. Rotate the interpupillary adjustment knob the full extent of its travel and measure the center distance at the eyepiece with a scale graduated in millimeters, record the reading. Next rotate the interpupillary adjustment knob in the opposite direction to the full extent of its travel and measure the center distance of the eyepiece, the difference in scale readings must be within the limits outlined in 3.8.

4.5.3.6 Magnification.- An appropriate method for determining magnification is to divide the diameter of the entrance pupil by the diameter of the exit pupil. The resulting value must fall within the limits of 3.10.

4.5.3.7 Cleanliness.- The telescopes shall be examined through the objective and eyepiece ends with the unaided eye. Inspection for moisture shall be made by the technique of shadowing. Inspection for dust particles shall be made by viewing a uniformly illuminated field having a brightness of 300 foot candles. Such illumination may be obtained by holding the sight two to three inches from a bench lamp containing two fifteen watt daylight fluorescent tubes with a white reflector. Dirt, moisture and foreign content shall not exceed the limits outlined in 3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8587675. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Sight Antioscillation XM55 is primarily intended for mounting in the Bell HU-1B Helicopter and is used by the gunner for selection of targets and observing the flight of the SS11 missile from time of launching from the helicopter until impact on target.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1)

(c) Applicable packaging data sheet number (see 5.1)

(d) Applicable stock number.

Custodian:

Army-MU

Preparing activity:

Army - MU

Project No. 1240-A341Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:46Z
Check the source to verify that this is the current version before use.